

NIKHITA GOWDA

U.S. Citizen | Columbus, OH | github.com/GowdaNikhita

Security engineer with about a year of hands-on experience in threat modeling, secure code review, and cloud security automation. I ship production Python and deploy infrastructure on AWS with Terraform. U.S. citizen, eligible for export-controlled work.

EXPERIENCE

Application Threat Intelligence Intern

2026 - Present

Keysight Technologies

- Automated BlackDuck software composition scans across the full dependency tree. Found 3 high-severity CVEs, flagged about 40% of dependencies, and handed engineering a ranked remediation backlog.
- Wired scan results into CI by correlating them with build-VM telemetry, which cut manual triage for the team by roughly 30%.
- Building STRIDE threat models for core application components. Mapping trust boundaries and abuse cases to guide secure-design reviews and prioritize the next round of fixes.

Web Development Intern

Aug - Sep 2024

Codsoft

- Built access-controlled REST APIs with role-based access control and server-side input validation across 5+ full-stack apps, closing broken-access-control and injection gaps.
- Separated authentication from business logic in a microservices job board and enforced authorization checks at each service boundary.

PROJECTS

SENTINEL: LLM Red-Teaming & Security Toolkit

Python, Anthropic API, Docker · 2025

- Ran 25 prompt-injection payloads across 5 categories against an LLM application. Measured a 29.1% overall success rate, peaking at 73% for data exfiltration.
- Threat-modeled the attack surface, then built and tested 3 mitigations: prompt hardening, input sanitization, and instruction-hierarchy tagging. Recommended layered defenses with output validation.
- Ingested 22 MITRE ATT&CK-mapped SIEM fixtures and ran 550 API calls with checkpointing to output structured attack-success metrics.

RetailOps / Kira: AIOps Incident-Response Platform

AWS, Terraform, Prometheus, Python · 2025

- Deployed a microservices stack on AWS with Terraform, ArgoCD, and GitHub Actions. Built Prometheus and Grafana dashboards for real-time alerting.
- Correlated CloudWatch logs, Prometheus metrics, and EKS cluster health to detect anomalies and surface root causes without manual triage.

BASTION: Network & Service Security Scanner

Python, nmap, NVD API, TLS · 2025

- Built a Python scanner that fingerprints services across a /24 subnet and maps them to known CVEs through the NVD API.
- Flagged weak TLS configurations and missing mutual TLS on internal endpoints in a deliberately vulnerable lab, then produced a prioritized fix list.

TECHNICAL SKILLS

Security: Threat modeling (STRIDE), secure code review, penetration testing, software composition analysis (BlackDuck), CVE triage, prompt injection, MITRE ATT&CK, SIEM/SOC, TLS/mTLS, CORS, authentication and access control

Languages: Python, C/C++, Java, JavaScript / Node.js, SQL

Cloud & Infrastructure: AWS (EC2, S3, Lambda, CloudWatch, EKS), Terraform, ArgoCD, Docker, Linux, Prometheus, Grafana, GitHub Actions

AI / ML: Anthropic API (tool-use), PyTorch, TensorFlow, Scikit-learn, Pydantic v2

EDUCATION

M.S. Computer Science & Engineering

Aug 2025 - May 2027

The Ohio State University

GPA 4.0 / 4.0

Coursework: Intro to Cybersecurity, Distributed Systems, Advanced Algorithms, Data Mining

B.E. Computer Science & Engineering

Aug 2021 - Jun 2025

JSS Science and Technology University

GPA 9.4 / 10

Coursework: Cybersecurity, Cryptography, Software Engineering